



Computer Organization & Assembly Language

Course: Computer Organization & Assembly Language (Lab) - CS-318

Lab-1

Introduction to EMU8086 and installation

"**Emu8086** is a powerful, offline and free software for emulation, disassembling and debugging of 8086 programs i.e., **16 bits/DOS.**" It is an Integrated Development Environment (IDE) that write a source, assemble it and link into .COM or .EXE file then trace it in machine code and source file. When we launch the Emu8086 Asm or Ist will start the assembler source editor. On the other hand, .exe and .obj files starts the disassembler and debugger units.

Let's move towards its installation:

Prerequisites:

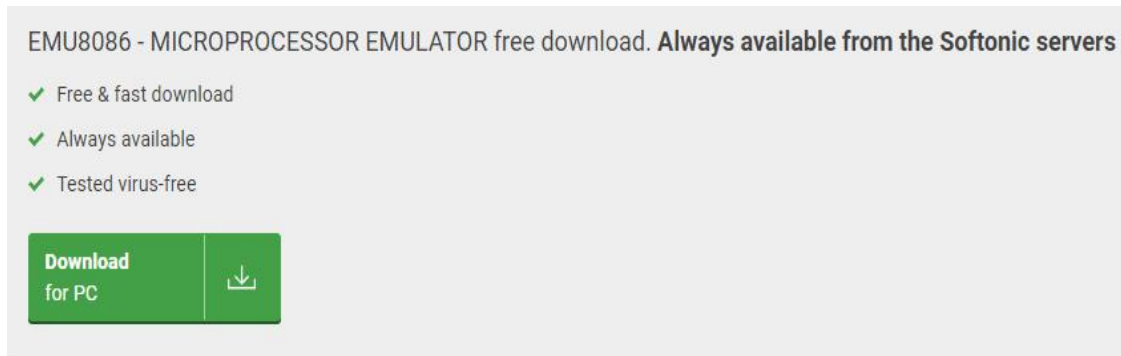
You must have

1. A PC
2. An active internet Connection
3. Administrative rights for Windows XP/Vista/7 users.
4. 10 Mb of hard disk space
5. 1024x768 or greater screen resolution.

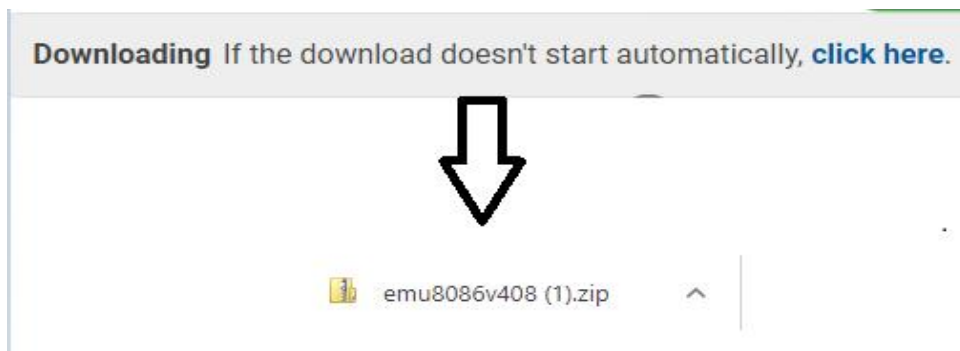
You can get Emu8086 through the link given below:

<https://emu8086-microprocessor-emulator.en.download.it/>

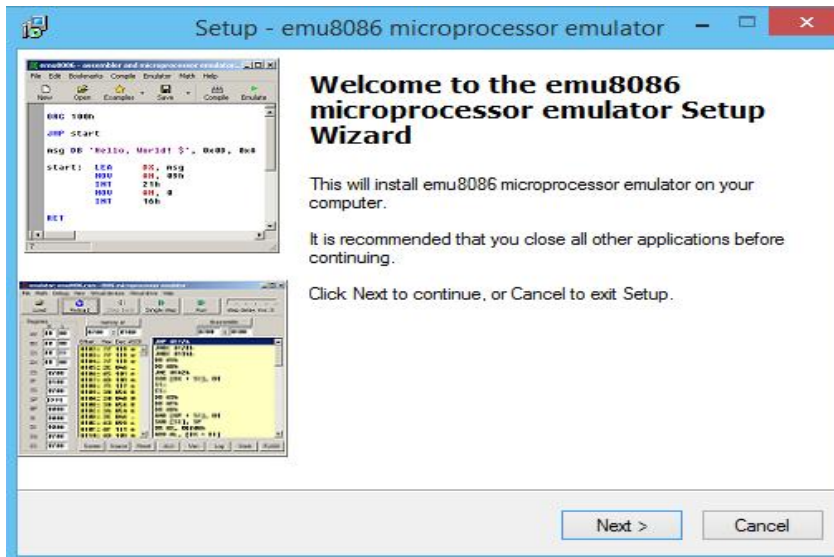
1. Click it and you will get a Website.



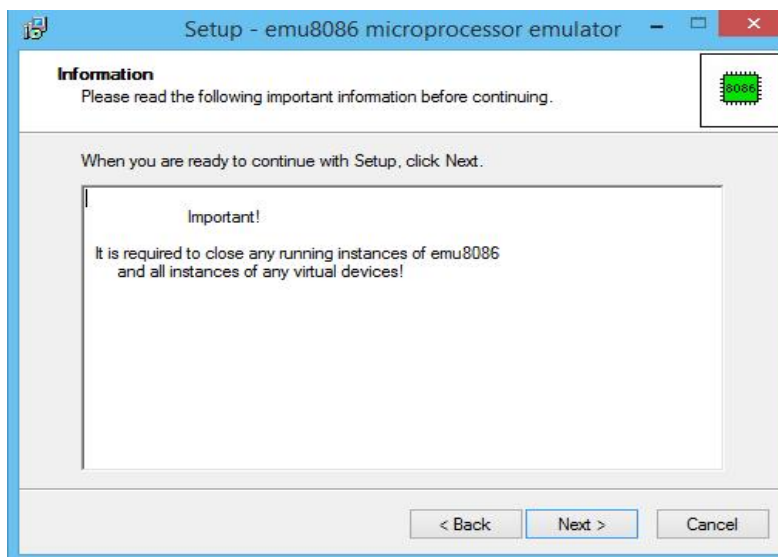
2. Click "Download for PC" button. This opens a new tab and the downloading starts. Observe that the zip file of 3Mb in your folders.



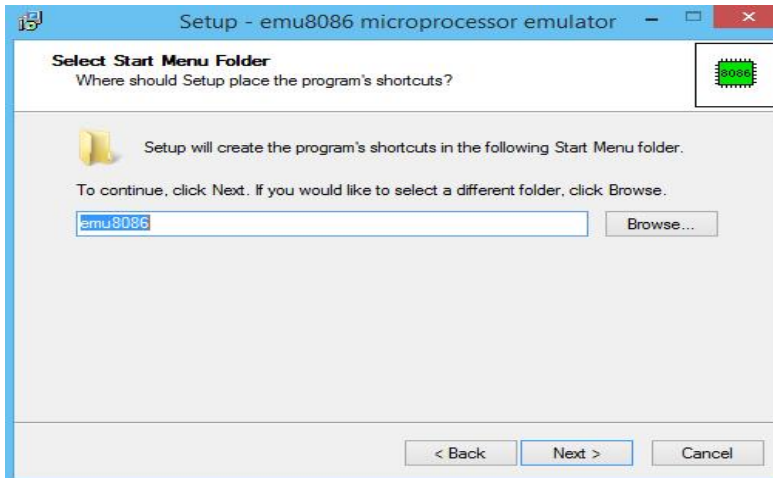
3. Click on the small arrow, open the folder. A folder will be pop up.
4. Double click at "Setup". It will ask for the permission. Allow it by clicking "yes".
The Window will appear that will start the setup process.



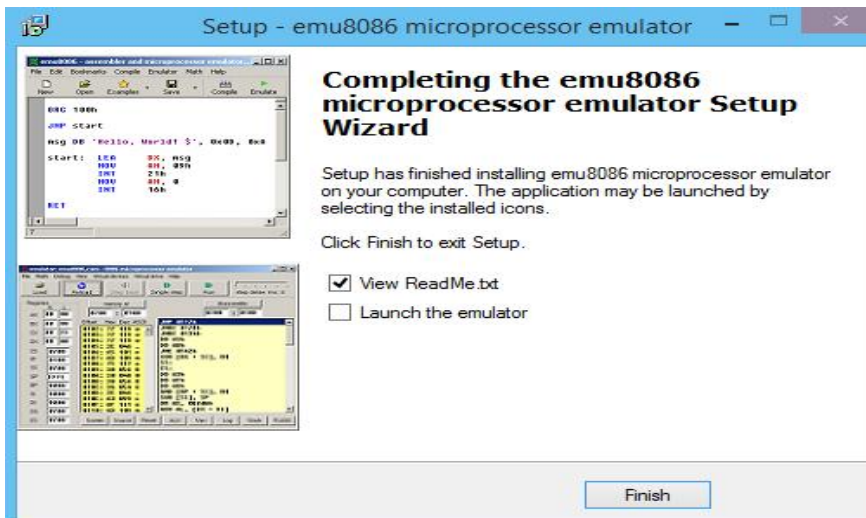
5. Click "Next" to continue the process. A new window will be appear like this:



6. Close All the Application running applications and click "Next".
7. Now, you have to give the path for the folder. By default the path is given for the **C** directory but you can change it by clicking **Browse**.



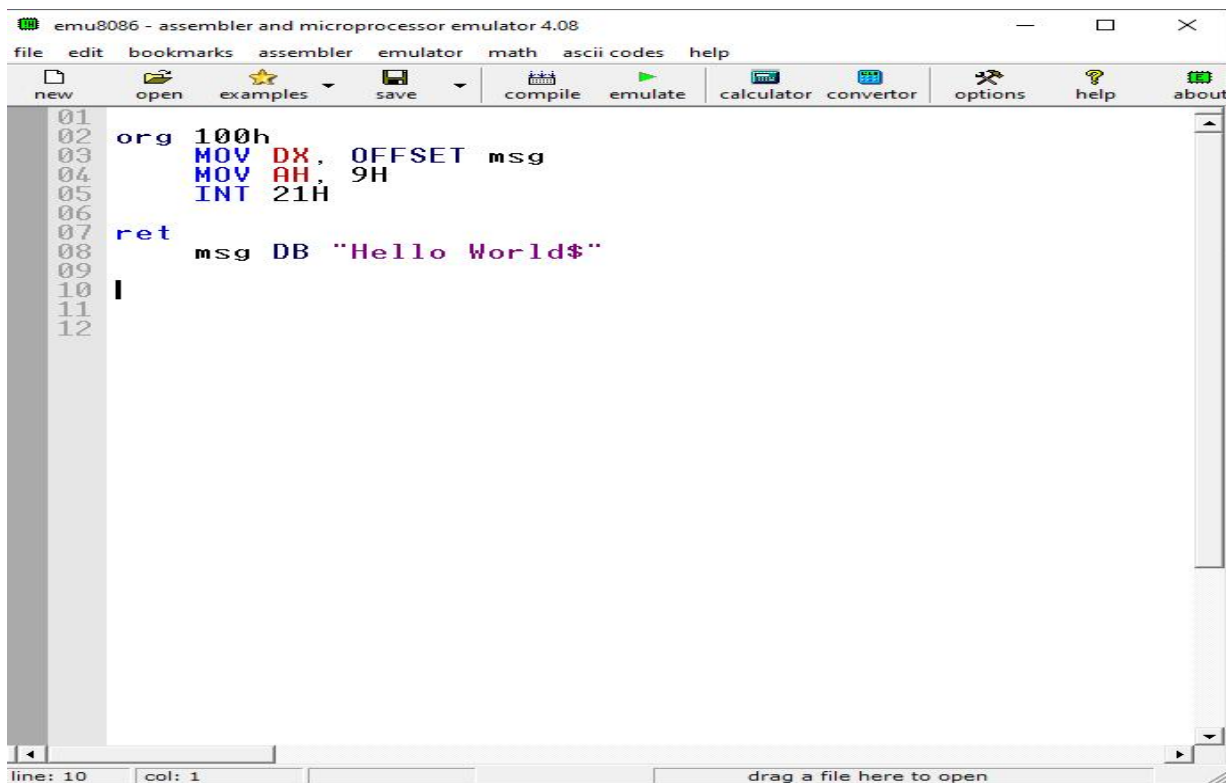
8. Read all the tabs appeared after it and then click "Next". In the last window, it will ask does we want to launch the Emu8086 or does we wish to read the instruction.
9. Choose the action you want to be launched or simply remove the tick mark and the setup is finished.



The Emu8086 icon will appear on your desktop. You can use it whenever you want.

Example: Hello World Program

- `org 100h`
→ Start the program at memory place 100h. (All small .COM programs begin here.)
- `MOV DX, OFFSET msg`
→ Put the address (location) of the text `msg` into register **DX**.
- `MOV AH, 9H`
→ Tell DOS: "I want to print a message." (Function number 9).
- `INT 21H`
→ Ask DOS to do the job. Here, it prints the message on the screen.
- `ret`
→ End the program and go back to DOS.
- `msg DB "Hello World$"`
→ This is the message stored in memory.
→ The `$` means "end of message" (so DOS knows where to stop printing).



The screenshot shows the emu8086 - assembler and microprocessor emulator 4.08 window. The interface includes a menu bar (file, edit, bookmarks, assembler, emulator, math, ascii codes, help) and a toolbar with icons for new, open, examples, save, compile, emulate, calculator, convertor, options, help, and about. The main text area displays the assembly code for the Hello World program, with line numbers 01 through 12 on the left. The code is as follows:

```
01  
02 org 100h  
03     MOV DX, OFFSET msg  
04     MOV AH, 9H  
05     INT 21H  
06  
07 ret  
08     msg DB "Hello World$"  
09  
10  
11  
12
```

The status bar at the bottom indicates "line: 10" and "col: 1". A message "drag a file here to open" is visible in the bottom right corner.

Output:

